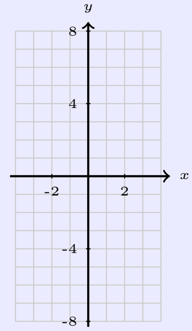
Cubic & Cube Root Inverses — Notes

Name: ____________________ Per: ______

Parent Functions & Key Points

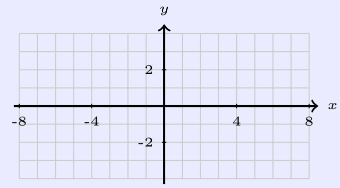
Cubic: $f(x) = x^3$

x	y
-2	
-1	
0	
1	
2	



Cube Root: $f(x) = \sqrt[3]{x}$

x	y
-8	
-1	
0	
1	
8	



Notice: The key points are ____________________ of each other! This is because they are ________________.

The 4-Step Inverse Method

1. Replace $f(x)$ with y
2. Swap x and y
3. Solve for y
4. Write using $f^{-1}(x)$ notation

Key Property: If (a, b) is on $f(x)$, then (b, a) is on $f^{-1}(x)$. Inverse functions reflect over the line $y = x$.

Example 1: $f(x) = (x + 2)^3 - 1$ *(Cubic)*

Step A: Identify $a =$ ____ $h =$ ____ $k =$ ____

Step B: Transform key points $(x, y) \to (x + h,\ ay + k)$

Parent	$f(x)$	Parent	$f(x)$
$(-2, -8)$	(,)	$(1, 1)$	(,)
$(-1, -1)$	(,)	$(2, 8)$	(,)
$(0, 0)$	(,)		

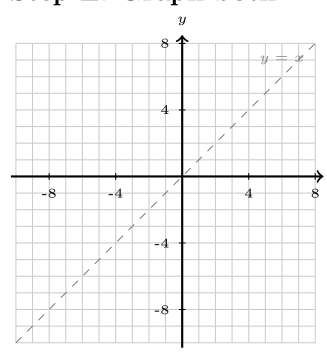
Step C: Find inverse (4-step)

$f^{-1}(x) =$ ______________________

Step D: Swap for inverse $(a, b) \to (b, a)$

$f(x)$	$f^{-1}(x)$
(,)	(,)
(,)	(,)
(,)	(,)

Step E: Graph both



Example 2: $f(x) = 2\sqrt[3]{x - 4} + 1$ *(Cube Root)*

Step A: Identify $a =$ ____ $h =$ ____ $k =$ ____

Step B: Transform key points $(x, y) \to (x + h,\ ay + k)$

Parent	$f(x)$	Parent	$f(x)$
$(-8, -2)$	(,)	$(1, 1)$	(,)
$(-1, -1)$	(,)	$(8, 2)$	(,)
$(0, 0)$	(,)		

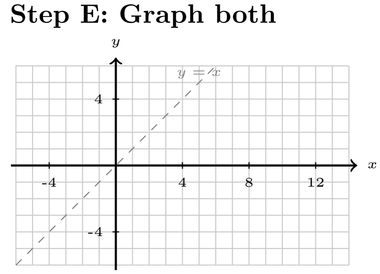
Step C: Find inverse (4-step)

$f^{-1}(x) =$ ______________________

Step D: Swap for inverse $(a, b) \to (b, a)$

$f(x)$	$f^{-1}(x)$
(,)	(,)
(,)	(,)
(,)	(,)

Step E: Graph both



Summary: Graphing a Function and Its Inverse

1. **Identify** a, h, k 2. **Transform** points: $(x, y) \to (x + h,\ ay + k)$ 3. **Graph** $f(x)$
2. **Find inverse** (4-step) 5. **Swap** $(a, b) \to (b, a)$ 6. **Graph** $f^{-1}(x)$ — reflects over $y = x$

Quick Reference: Attributes

Attribute	**Cubic:** $f(x) = x^3$	**Cube Root:** $f(x) = \sqrt[3]{x}$
Domain / Range	$(-\infty, \infty)$ / $(-\infty, \infty)$	$(-\infty, \infty)$ / $(-\infty, \infty)$
Inflection Point	$(0, 0)$	$(0, 0)$
End Behavior	$x \to \pm\infty \Rightarrow y \to \pm\infty$	Same
Symmetry	Odd (origin symmetry)	Odd (origin symmetry)